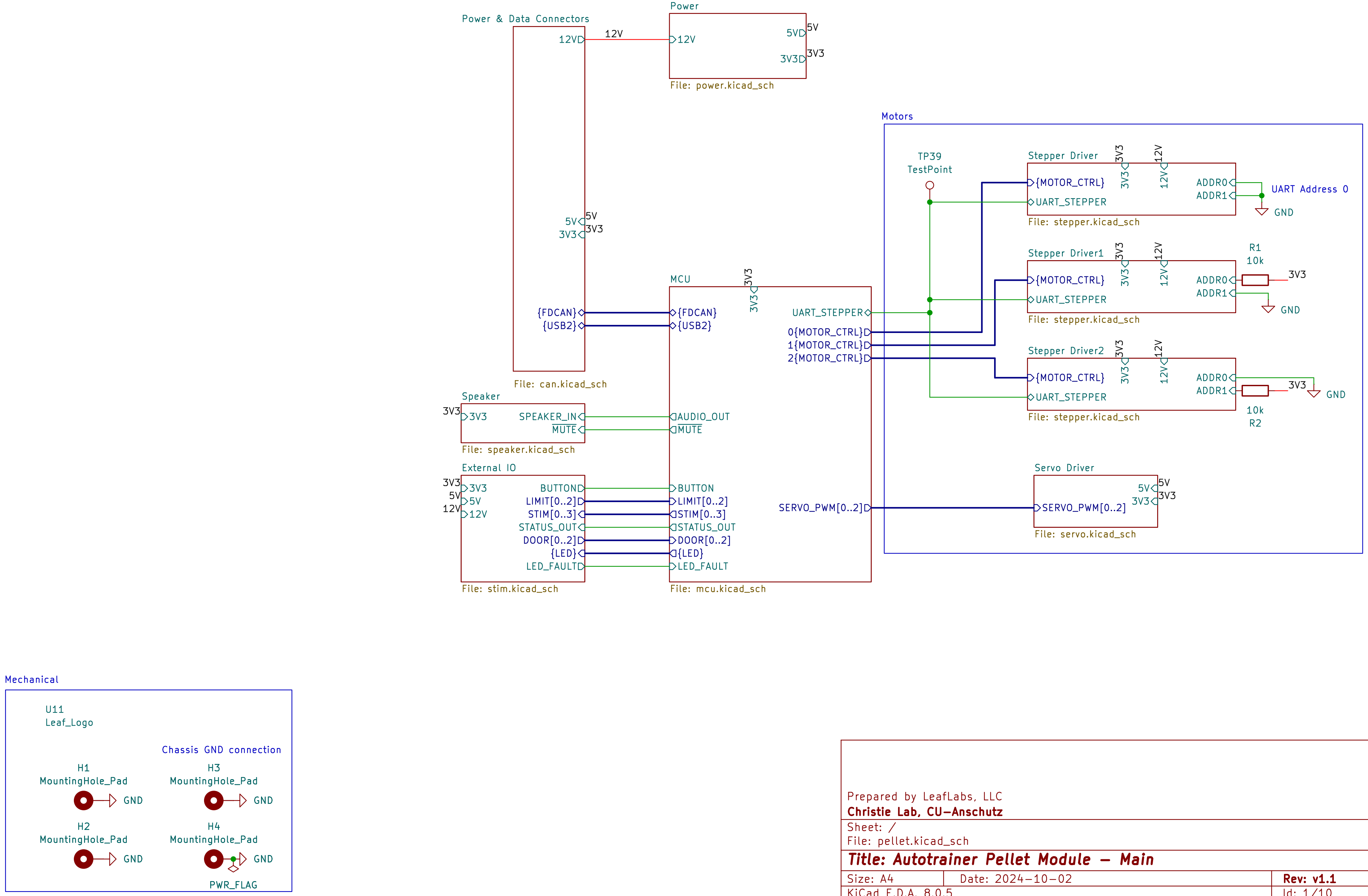
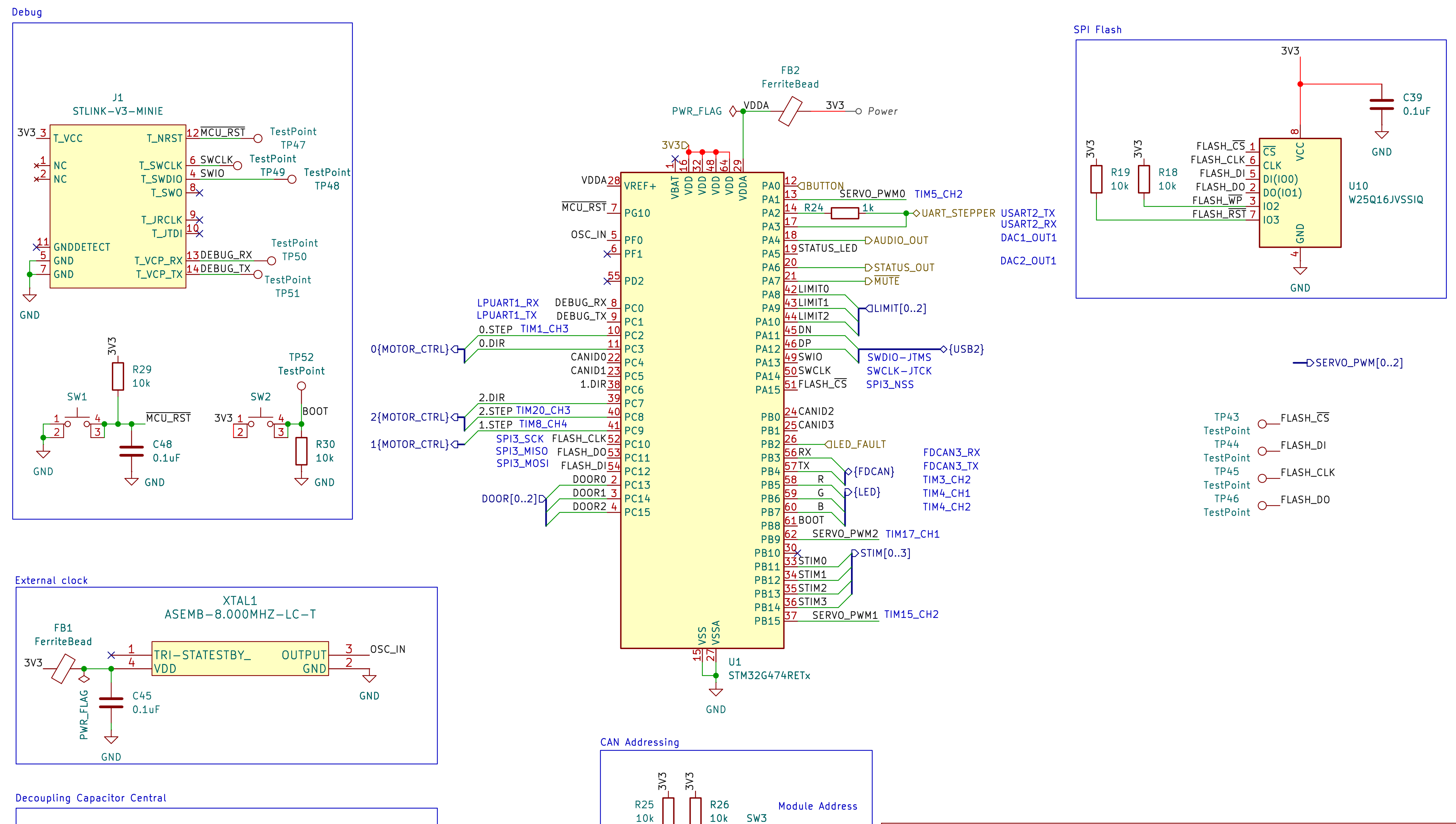


Prepared by LeafLabs, LLC
Christie Lab, CU-Anschutz

Sheet: /
File: pellet.kicad_sch

Title: Autotrainer Pellet Module - Main

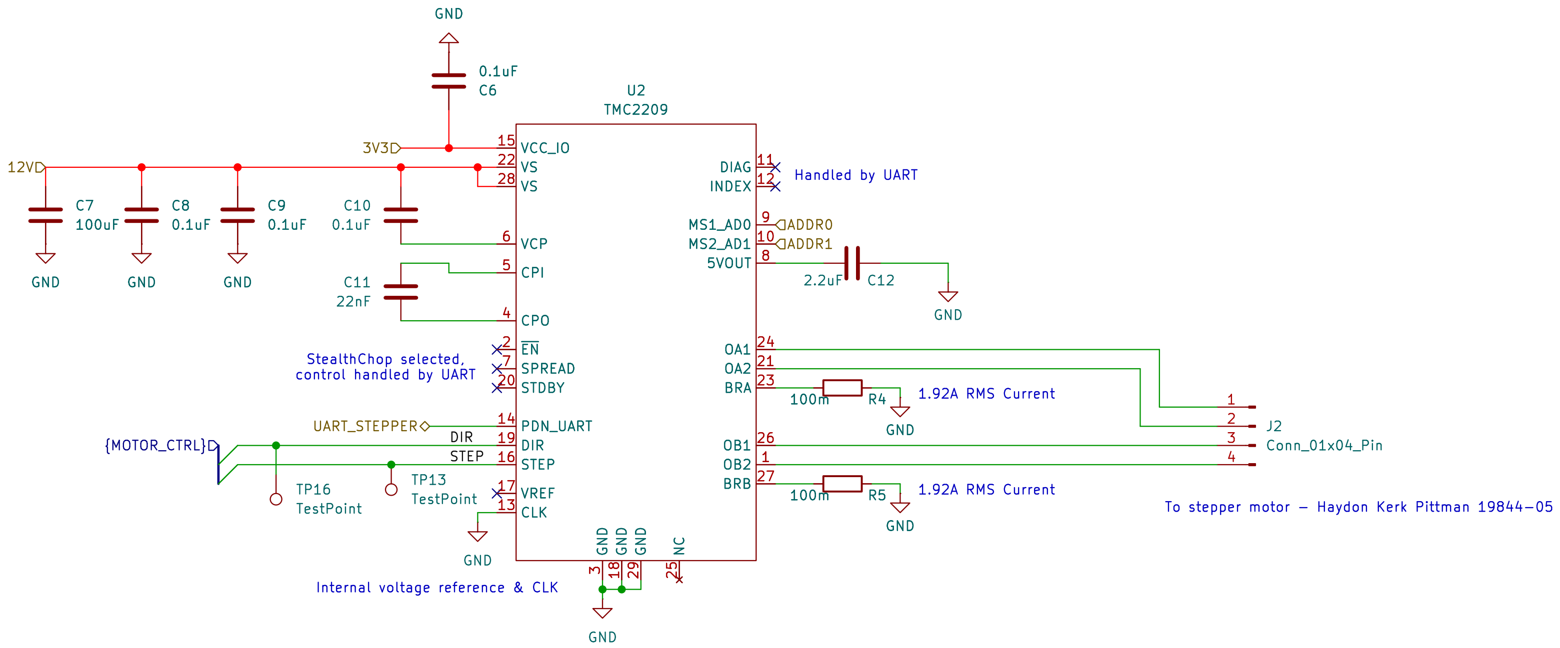
Size: A4 Date: 2024-10-02 **Rev: v1.1**
KiCad E.D.A. 8.0.5 Id: 1/10



Prepared by LeafLabs, LLC
Christie Lab, CU-Anschutz
Sheet: /MCU/
File: mcu.kicad_sch

Title: Autotrainer Pellet Module - MCU

Size: A4	Date: 2024-10-02	**Rev: v1.1**
KiCad E.D.A. 8.0.5		Id: 2/10



Prepared by LeafLabs, LLC

Christie Lab, CU-Anschutz

Sheet: /Stepper Driver/

File: stepper.kicad_sch

Title: Autotrainer Pellet Module - Stepper

Size: A4	Date: 2024-10-02	**Rev: v1.1**
KiCad E.D.A. 8.0.5		Id: 3/10

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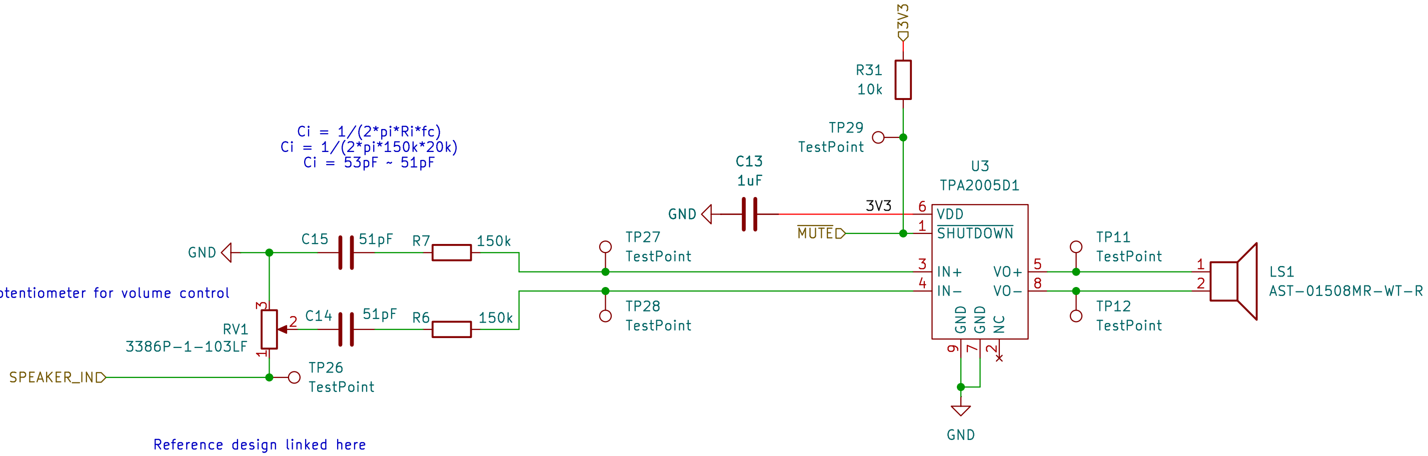
Sheet: /Servo Driver/

File: servo.kicad_sch

Title: Autotrainer Pellet Module – Servo

Size: A4	Date: 2024-10-02	**Rev: v1.1**
KiCad E.D.A. 8.0.5		Id: 4/10



Mouse audible range is 1k−91kHz
Human audible range is 20Hz−20kHz

Design for audio between 1k−20kHz

Prepared by LeafLabs, LLC

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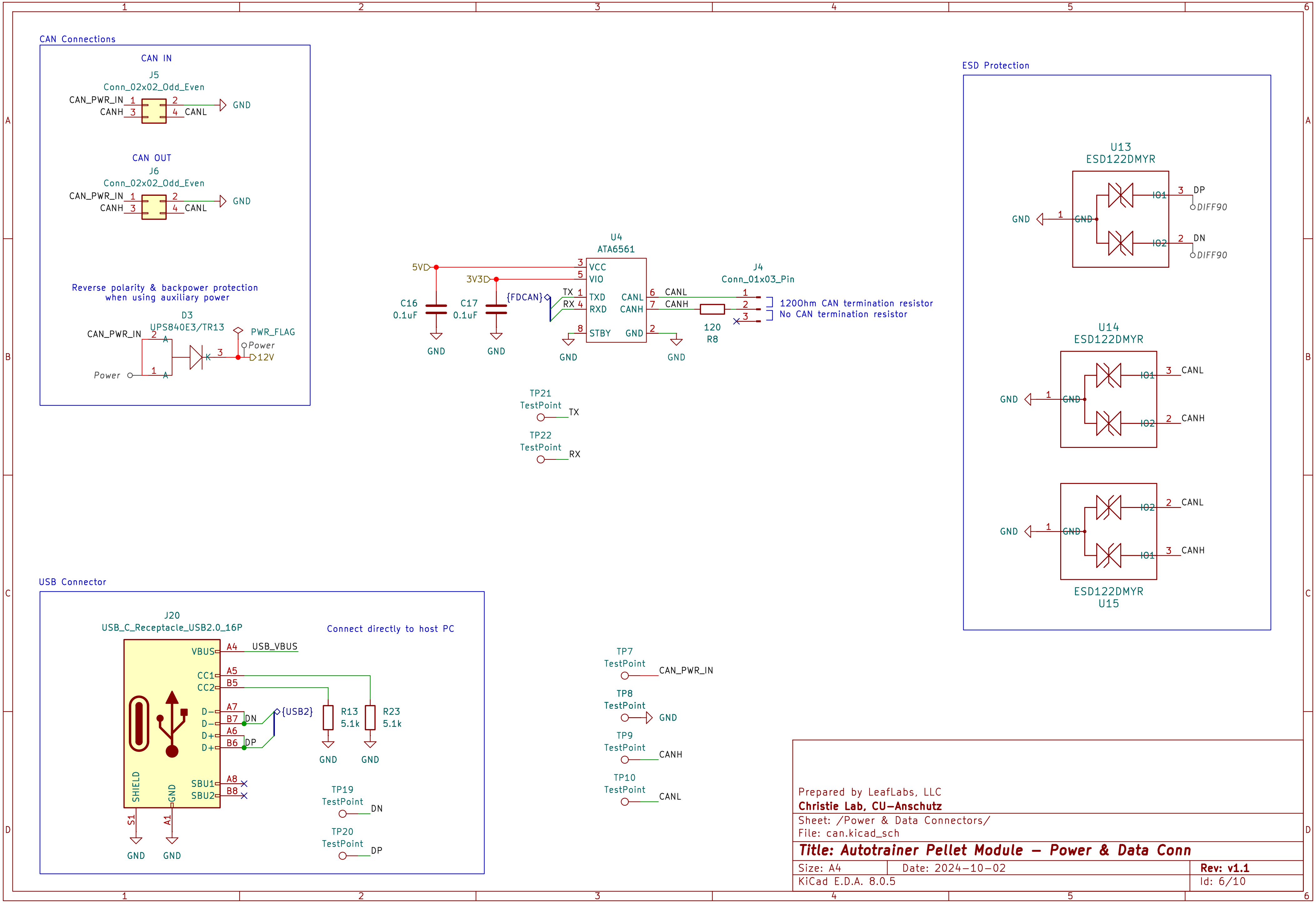
Sheet: /Speaker/

File: speaker.kicad_sch

Title: Autotrainer Pellet Module − Speaker

Size: A4	Date: 2024−10−02	**Rev: v1.1**
KiCad E.D.A. 8.0.5		Id: 5/10



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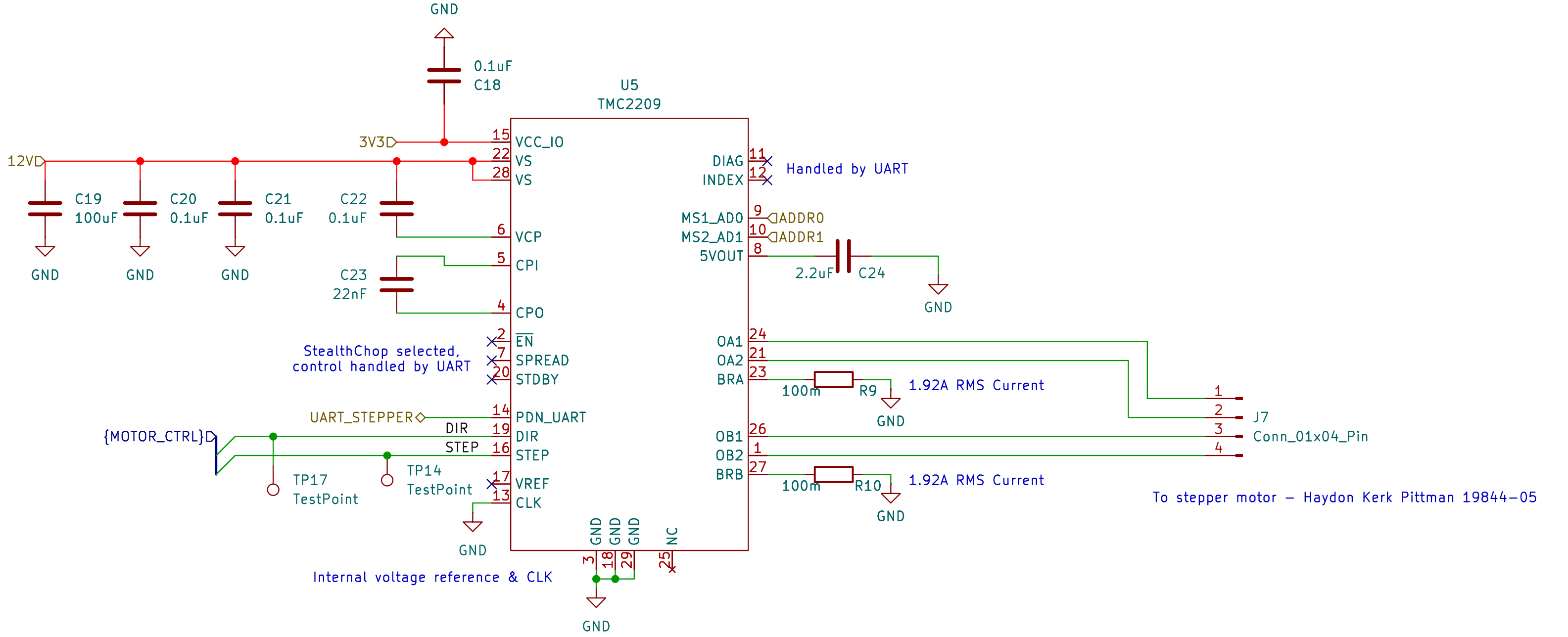
Christie Lab, CU-Anschutz

Sheet: /Power & Data Connectors/
File: can.kicad_sch

Title: Autotrainer Pellet Module — Power & Data Conn

Size: A4 | Date: 2024-10-02 | **Rev: v1.1**
KiCad E.D.A. 8.0.5 | Id: 6/10



Prepared by LeafLabs, LLC

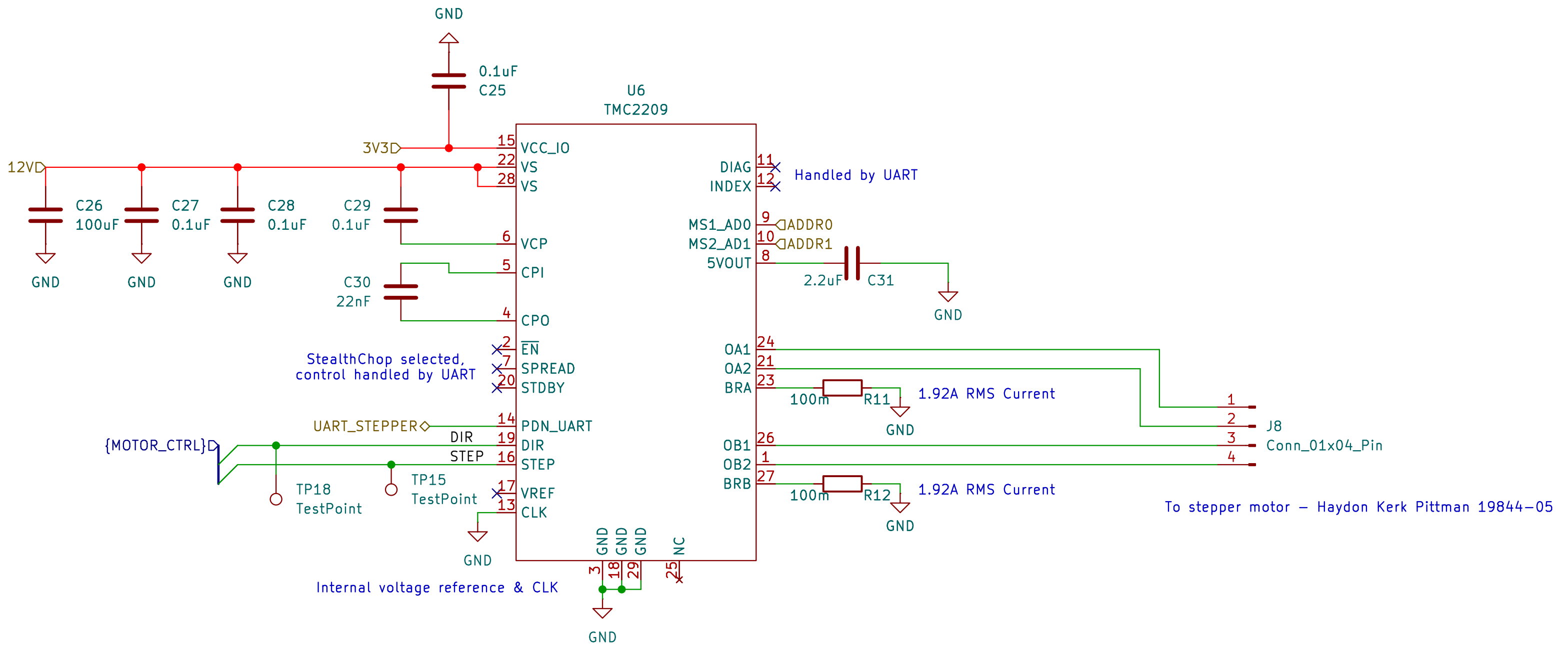
Christie Lab, CU-Anschutz

Sheet: /Stepper Driver1/

File: stepper.kicad_sch

Title: Autotrainer Pellet Module – Stepper

Size: A4	Date: 2024-10-02	**Rev: v1.1**
KiCad E.D.A. 8.0.5		Id: 7/10



Prepared by LeafLabs, LLC

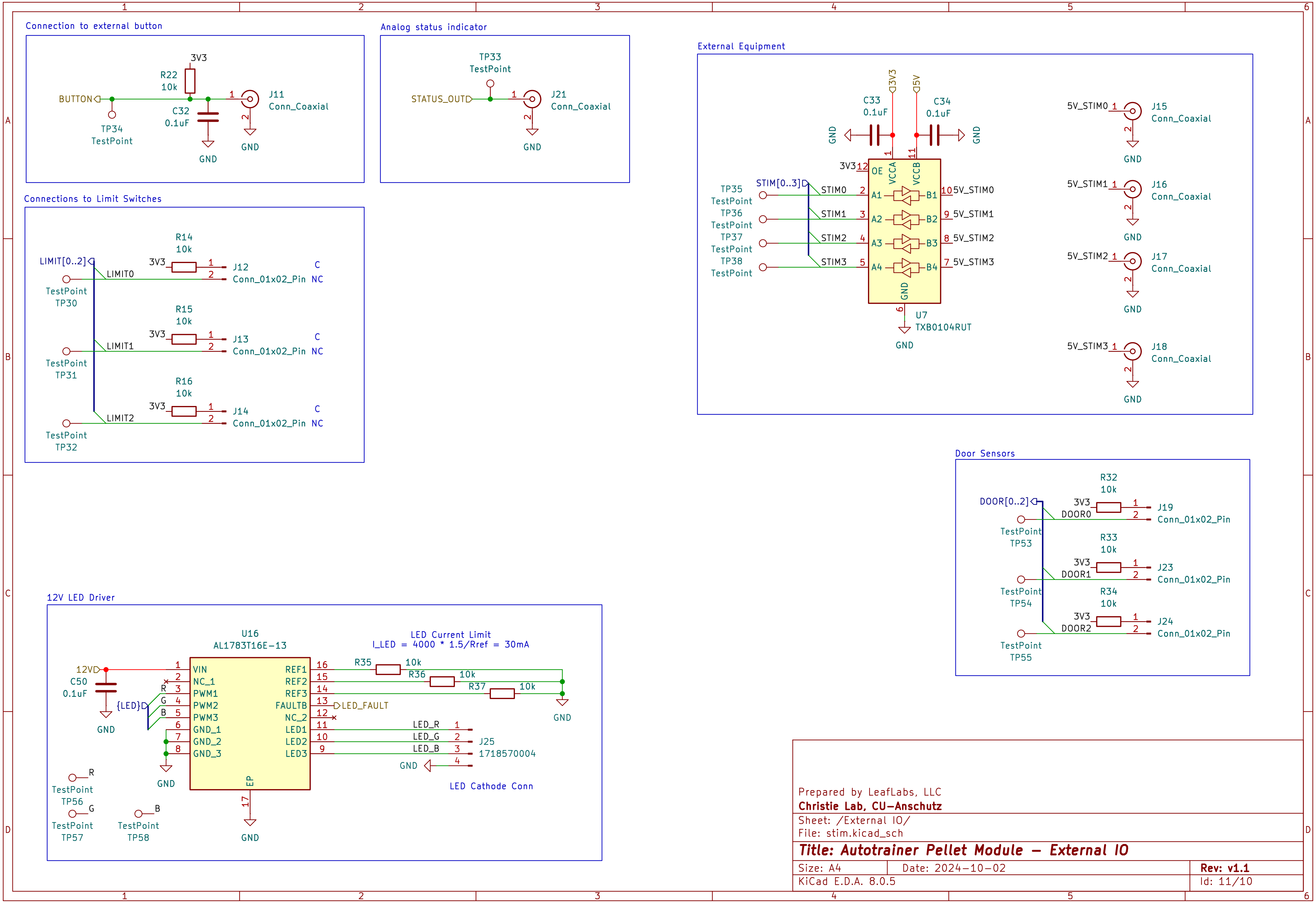
Christie Lab, CU–Anschutz

Sheet: /Stepper Driver2/

File: stepper.kicad_sch

Title: Autotrainer Pellet Module – Stepper

Size: A4	Date: 2024–10–02	**Rev: v1.1**
KiCad E.D.A. 8.0.5		Id: 8/10

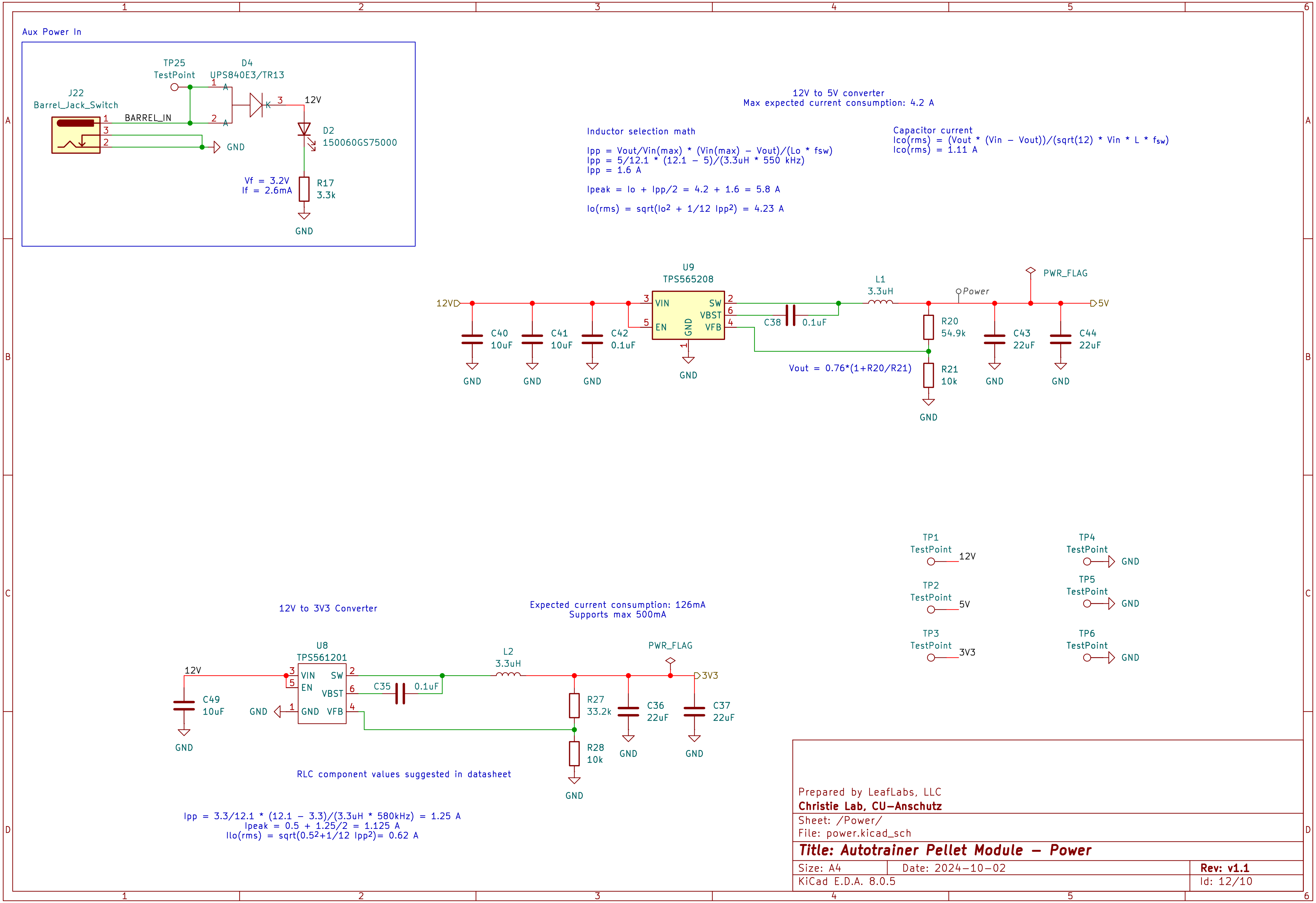


Prepared by LeafLabs, LLC
Christie Lab, CU-Anschutz

Sheet: /External IO/
File: stim.kicad_sch

Title: Autotrainer Pellet Module - External IO

Size: A4 Date: 2024-10-02 **Rev: v1.1**
KiCad E.D.A. 8.0.5 Id: 11/10


Aux Power In
TP25
TestPoint
D4
UPS840E3/TR13
J22
Barrel_Jack_Switch
BARREL_IN
12V
D2
150060GS75000
GND
Vf = 3.2V
If = 2.6mA
R17
3.3k
GND
12V to 5V converter
Max expected current consumption: 4.2 A
Inductor selection math
Ipp = Vout/Vin(max) * (Vin(max) − Vout)/(Lo * fsw)
Ipp = 5/12.1 * (12.1 − 5)/(3.3uH * 550 kHz)
Ipp = 1.6 A
Ipeak = Io + Ipp/2 = 4.2 + 1.6 = 5.8 A
Io(rms) = sqrt(Io2 + 1/12 Ipp2) = 4.23 A
Capacitor current
Ico(rms) = (Vout * (Vin − Vout))/(sqrt(12) * Vin * L * fsw)
Ico(rms) = 1.11 A
U9
TPS565208
VIN
EN
GND
SW
VBST
VFB
12V
C40
10uF
C41
10uF
C42
0.1uF
C38
0.1uF
L1
3.3uH
Power
PWR_FLAG
5V
R20
54.9k
R21
10k
C43
22uF
C44
22uF
Vout = 0.76*(1+R20/R21)
GND
TP1
TestPoint
12V
TP2
TestPoint
5V
TP3
TestPoint
3V3
TP4
TestPoint
GND
TP5
TestPoint
GND
TP6
TestPoint
GND
12V to 3V3 Converter
Expected current consumption: 126mA
Supports max 500mA
U8
TPS561201
VIN
EN
GND
SW
VBST
VFB
12V
C49
10uF
C35
0.1uF
L2
3.3uH
PWR_FLAG
3V3
R27
33.2k
R28
10k
C36
22uF
C37
22uF
GND
RLC component values suggested in datasheet
Ipp = 3.3/12.1 * (12.1 − 3.3)/(3.3uH * 580kHz) = 1.25 A
Ipeak = 0.5 + 1.25/2 = 1.125 A
Ilo(rms) = sqrt(0.52+1/12 Ipp2)= 0.62 A
Prepared by LeafLabs, LLC
Christie Lab, CU-Anschutz
Sheet: /Power/
File: power.kicad_sch
Title: Autotrainer Pellet Module − Power
Size: A4
Date: 2024-10-02
KiCad E.D.A. 8.0.5
Rev: v1.1
Id: 12/10